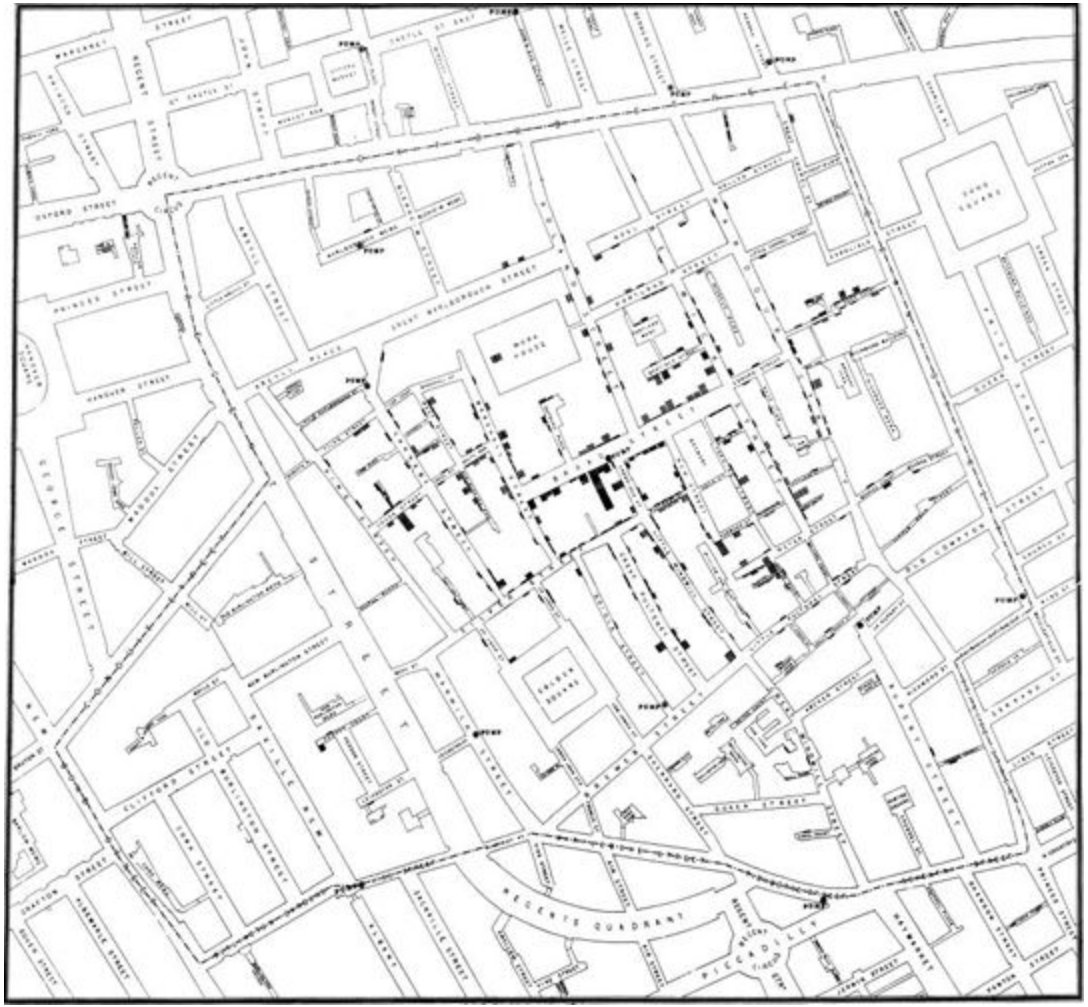
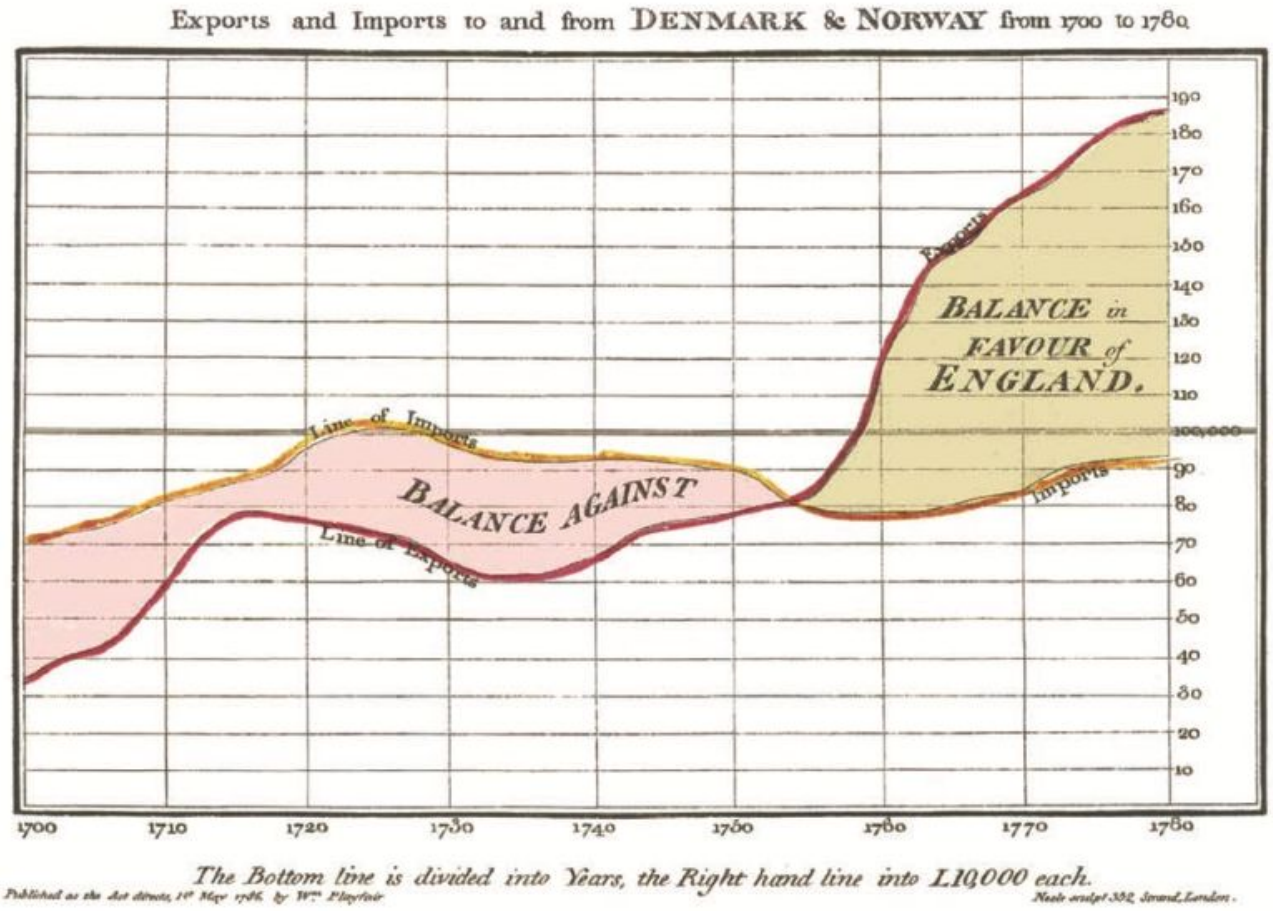
Crafting the Data Narrative and Storytelling

BI304

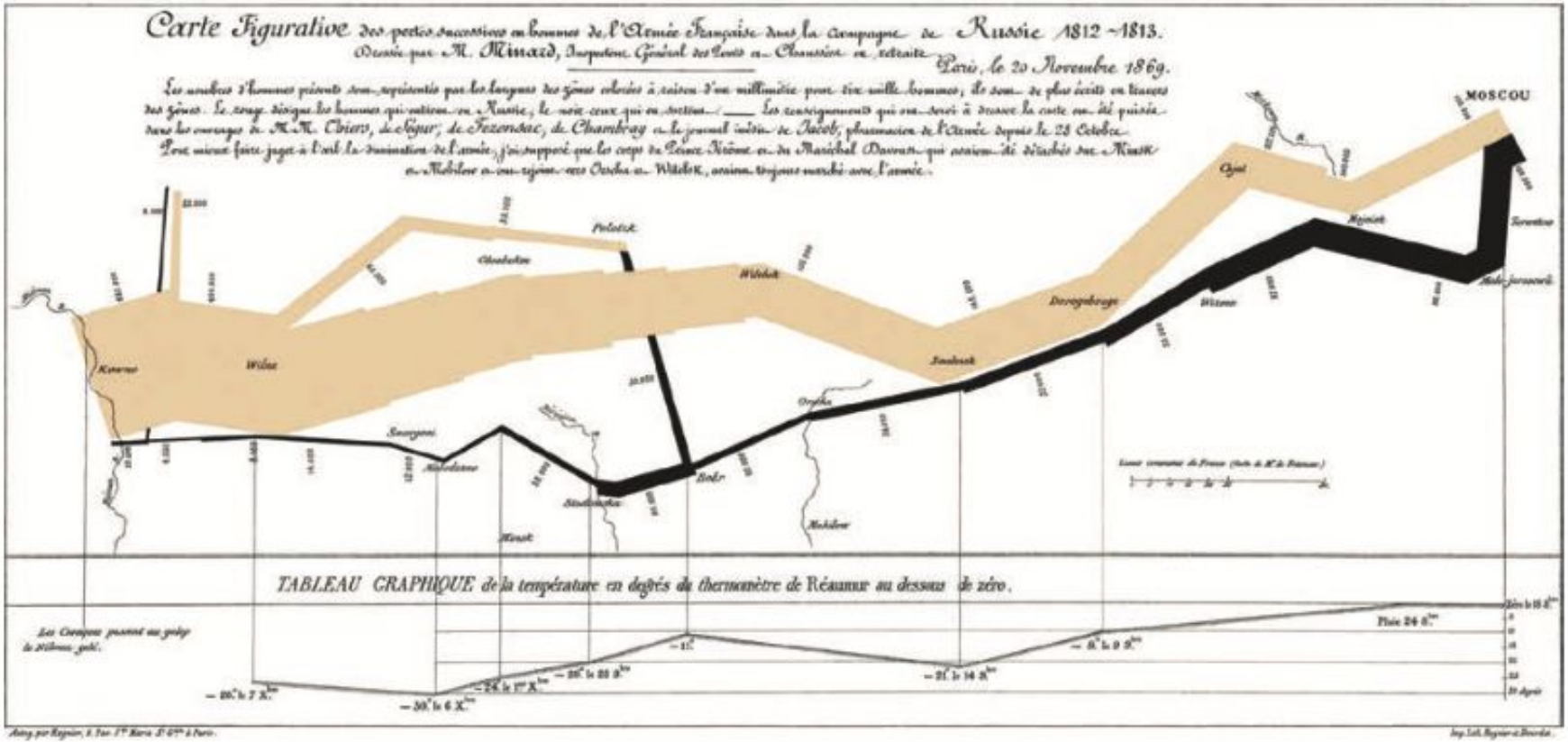
Noptanit Chotisarn
Thammasat Business School



https://en.wikipedia.org/wiki/The_Ghost_Map



The First Time-Series Line Chart Created Commercial and Political Atlas by William PlayFair in 1801.



Decimation of Napoleon's Army during the 1812 Russian Campaign.

Storytelling

- Visual or not visual, automated or manual, online or paper based, business reporting is not much different than telling a story
- Stories bring life to data and facts. They can help you make sense and order out of a disparate collection of facts. They make it easier to remember key points and can paint a vivid picture of what the future can look like. Stories also create interactivity—people put themselves into stories and can relate to the situation
- Most important, storytelling is interactive—typically, the presenter uses words and pictures that audience members can put themselves into. As a result, they become more engaged and better understand the information

Think of Your Analysis as a Story

When crafting a data-rich story, the first objective is to find the story.

- Who are the characters?
- What is the drama or challenge?
- What hurdles have to be overcome?

And at the end of your story,

- what do you want your audience to do as a result?

Storyboard

- With the story elements in place, write out the storyboard, which represents the structure and form of your story
- Although it’s tempting to skip this step, it is better first to understand the story you are telling and then to focus on the presentation structure and form.
- Once the storyboard is in place, the other elements will fall into place
- The storyboard will help you think about the best analogies or metaphors, clearly set up challenge or opportunity, and finally see the flow and transitions needed
- The storyboard also helps you focus on key visuals (graphs, charts, and graphics) that you need your executives to recall

Data Comics

https://datacomics.github.io/

https://www.apt.ch/sites/default/files/publications/datacomic-15022021.pdf

https://aviz.fr/~bbach/datacomics/

Information Dashboards

- Information dashboards are common components of most, if not all, BI or business analytics platforms, business performance management systems, and performance measurement software suites
- Dashboards provide visual displays of important information that is consolidated and arranged on a single screen so that the information can be digested at a single glance and easily drilled in and further explored

Dashboard Design

According to Eckerson (2006), a well-known expert on BI in general and dashboards in particular, the most distinctive feature of a dashboard is its three layers of information:

1. Monitoring: Graphical, abstracted data to monitor key performance metrics
2. Analysis: Summarized dimensional data to analyze the root cause of problems
3. Management: Detailed operational data that identify what actions to take to resolve a problem

Tufte’s Rules Rule

"The goal of information design is to convey the greatest number of ideas in the shortest time with the least ink in the smallest space."

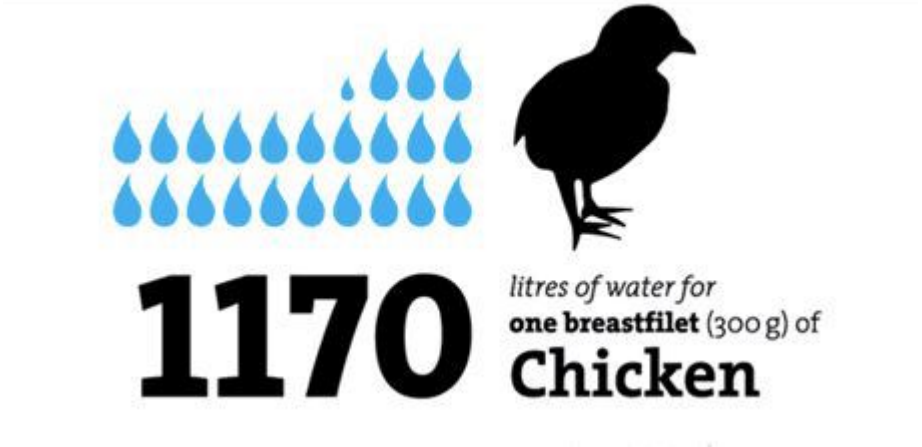
--- Edward Tufe’s The Visual Display of Quantitative Information

Tufte’s Rules Rule

- Rule 1: Show data.
- Rule 2: Use labels (but not too many).
- Rule 3: Avoid chartjunk.
- Rule 4: Maximize data-ink ratio.

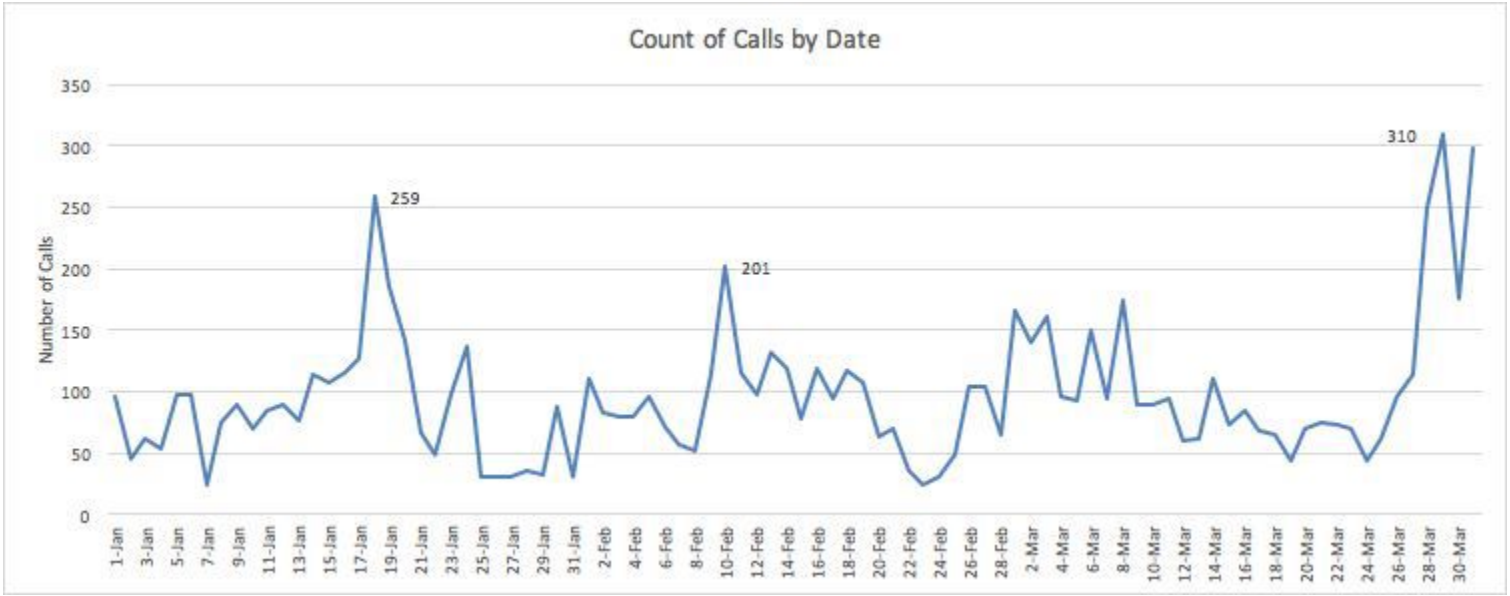
Rule No.1: Show Data

- No data visualization is complete without data. That is, after all, what they're meant to represent. What is the graphic below telling us?
- In this example from The Virtual Water Project, the graphic and the data measurement work together to provide useful information about the water footprint of chicken.



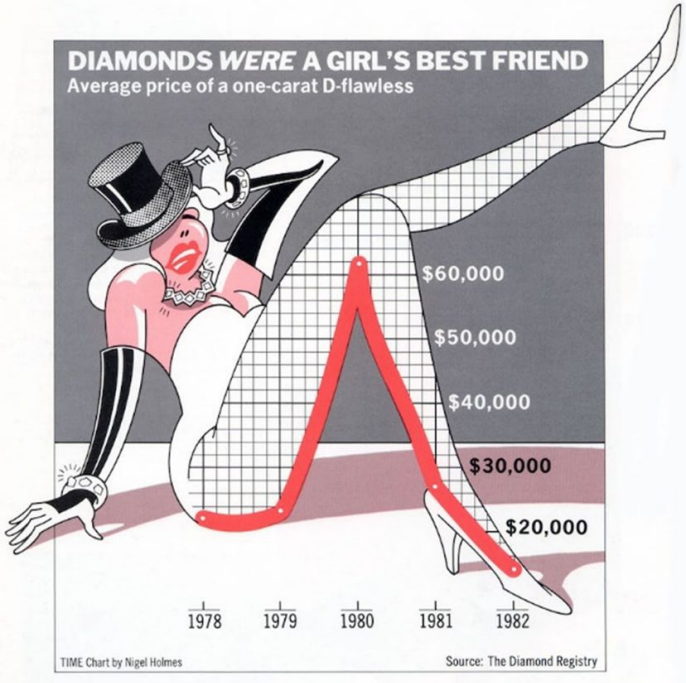
Rule No.2: Use Labels

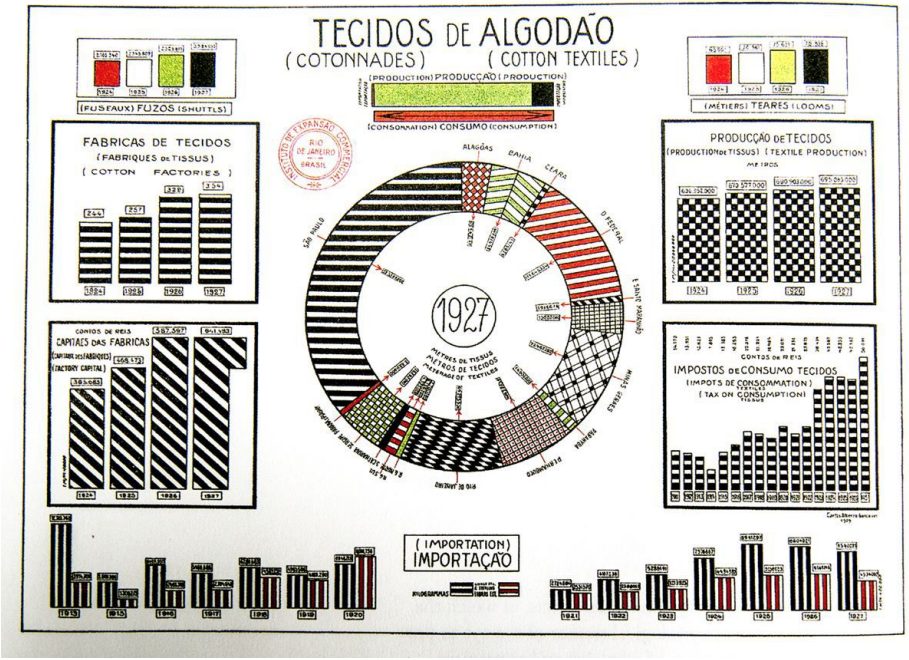
- Labels bring trustworthy to data. Data Integrity is important.
- An excellent way to include data in your visualization is through labeling.



Rule No.3: Avoid Chartjunk

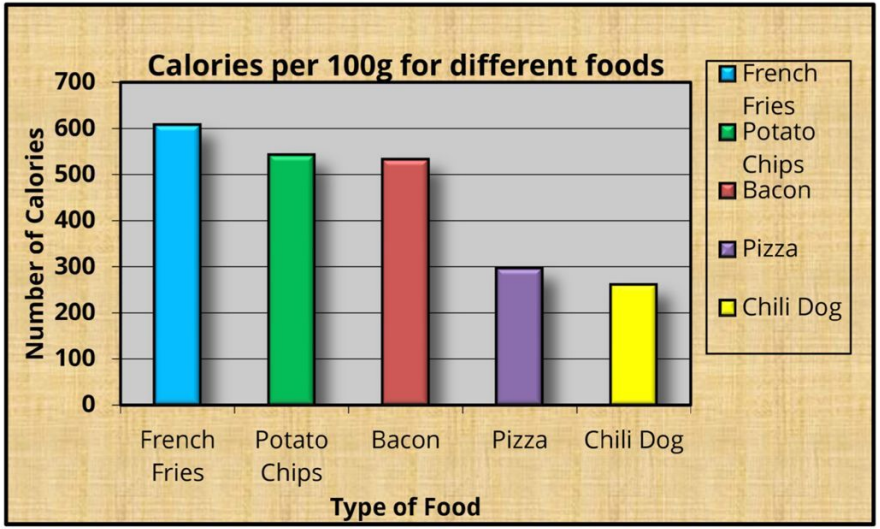
- Unintended optical art (such as lines and patterns in chart)
- The grid



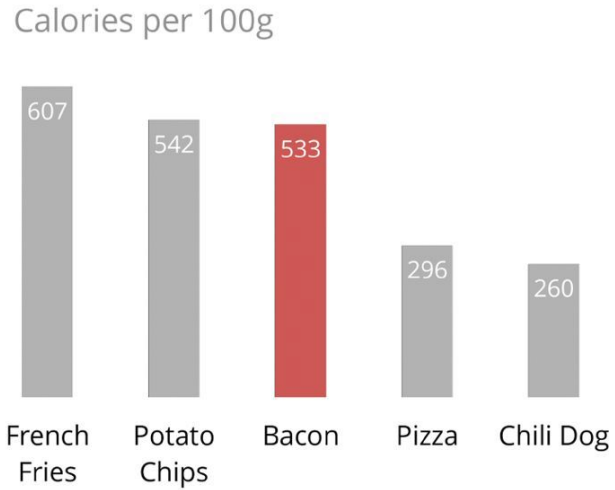


Rule No.4: Maximize Data-Ink Ratio

$$\text{Data-Ink Ratio} = \frac{\text{Data ink}}{\text{Total ink used in graphic}}$$



Remove
to improve
(the **data-ink** ratio)



Visualization caution

The Lie Factor

3d+ Scatter Plot (Avoid It)

1d Bar Chart (Don’t Do)

Trustworthy In Bar Chart

Aspect Ratio

Line Graph: Poor Examples

The Hate Of Pie Chart

Avoid Rainbow Map

Think About The Color Blind

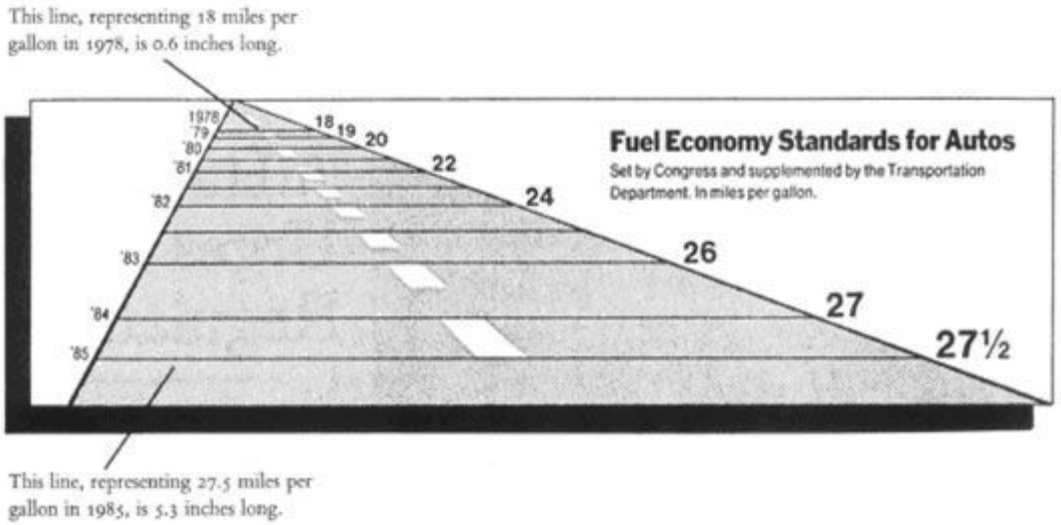
The size of an effect shown in the graphic
The size of the effect actually in the data?

The Lie Factor

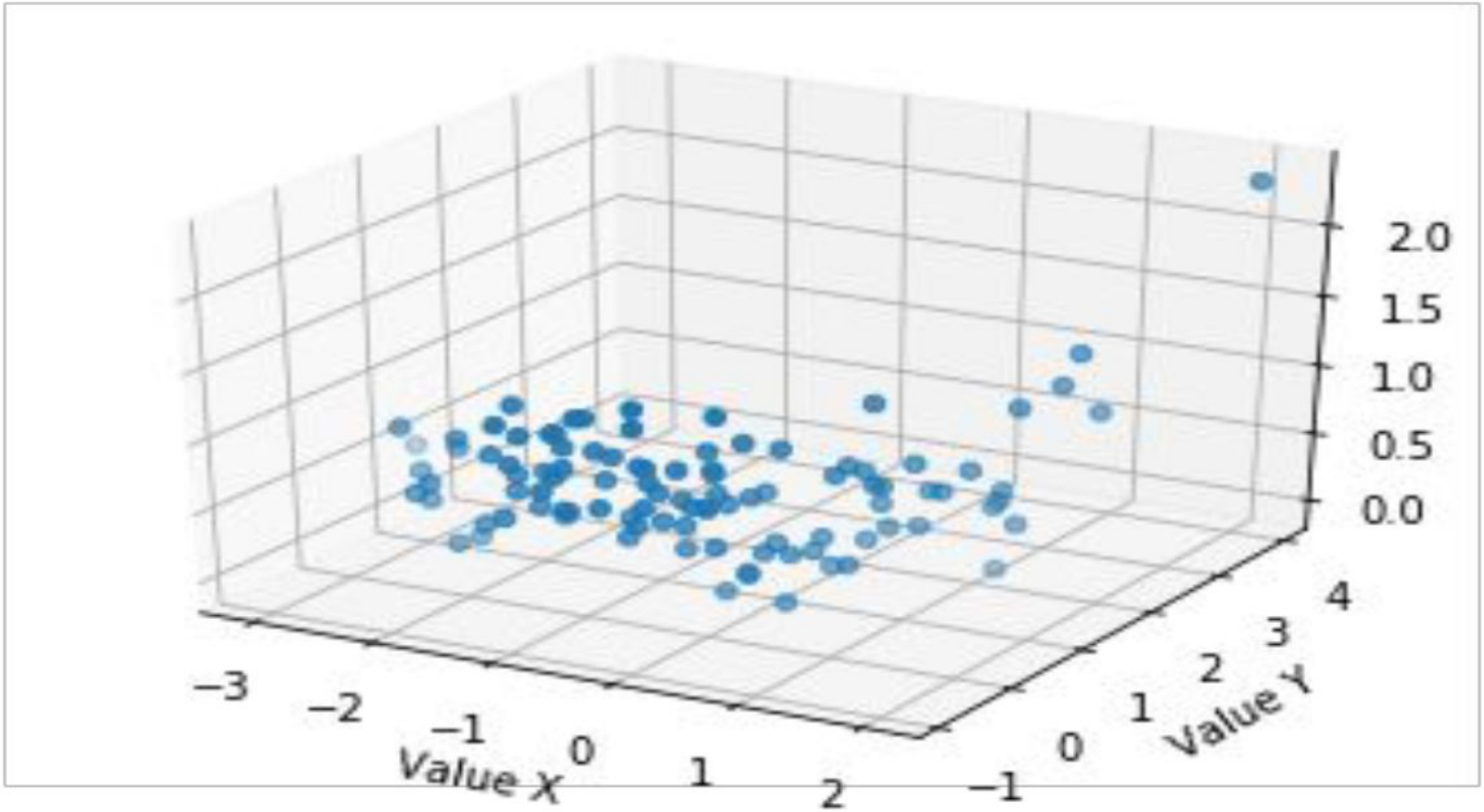
- The “Lie Factor” is a value to describe the relation between the size of effect shown in a graphic and the size of effect shown in the data.

Example:

- The standard required an increase in mileage from 18 to 27.5, an increase of 53%. The magnitude of increase shown in the graph is 783%, which results in a lie factor of 14.8!

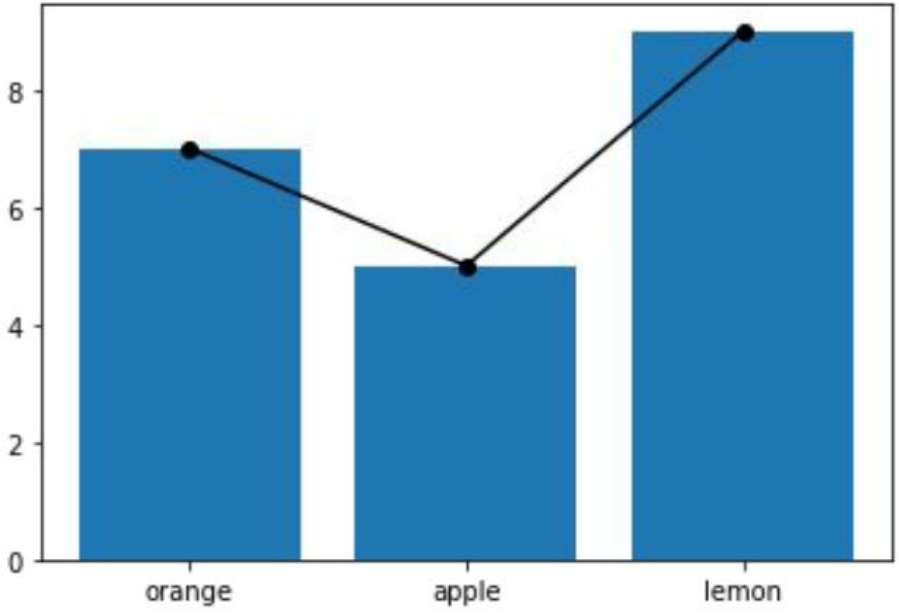


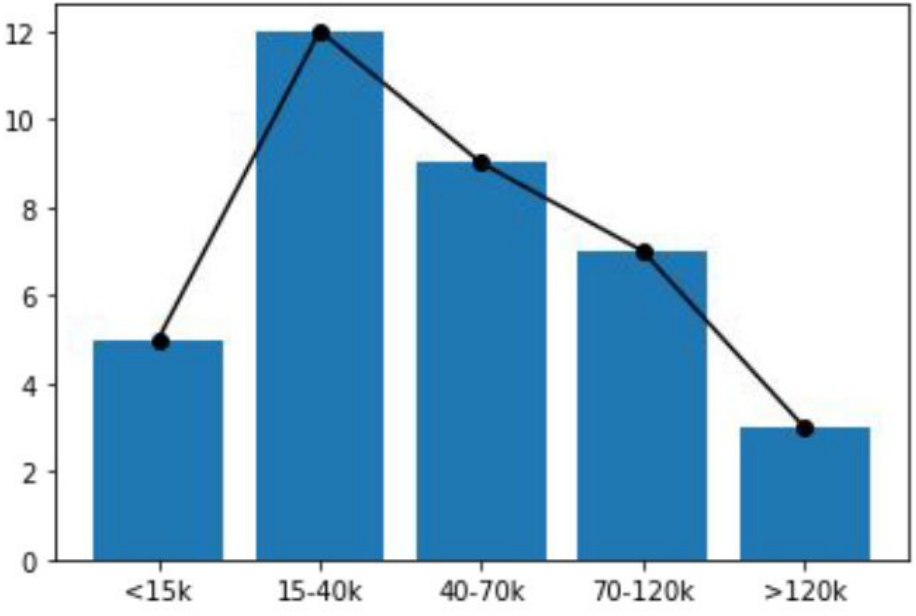
3d+ Scatter Plot (Avoid It)



1d Bar Chart (Don’t Do)

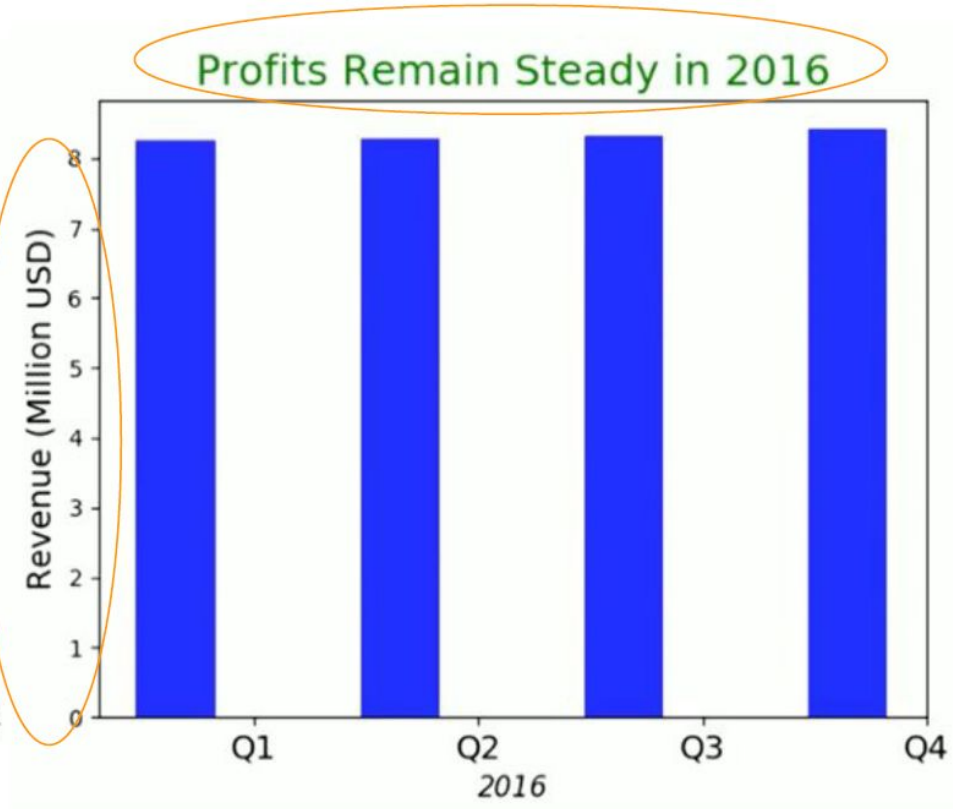
Don’t connect the bars with a line if it does not add information





Trustworthy In Bar Chart





Kirk. Data Visualisation: A Handbook for Data Driven Design. SAGE publications, 2016.

Aspect Ratio

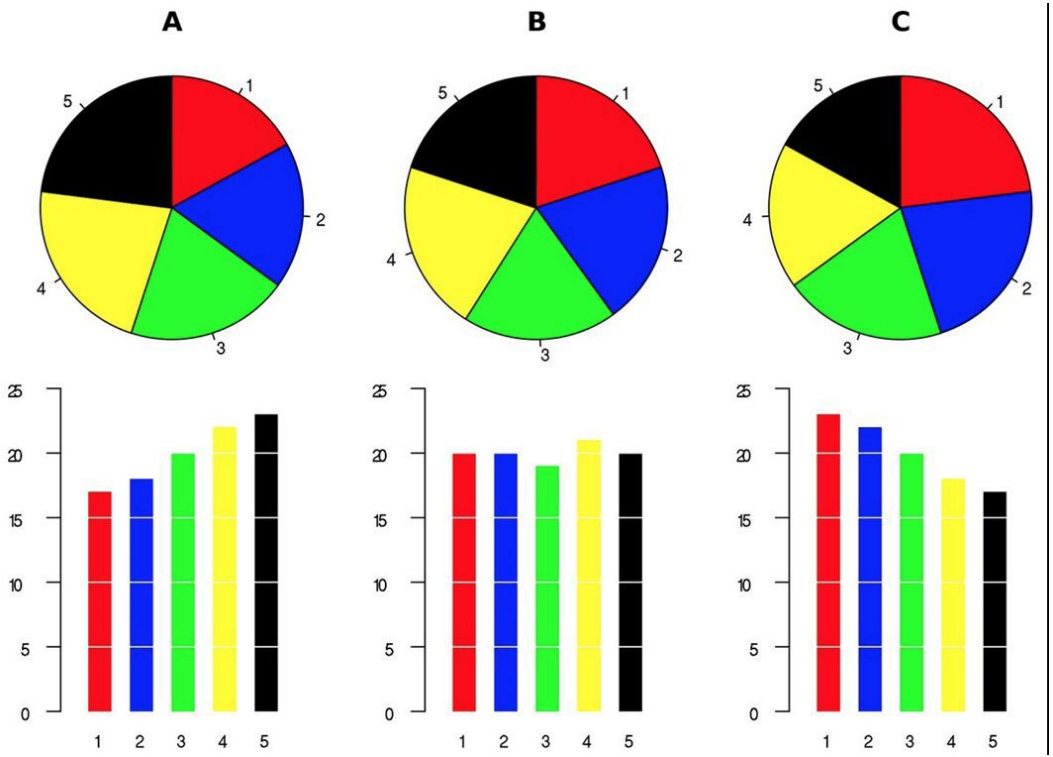


https://eagereyes.org/basics/banking-45-degrees

Line Graph: Poor Examples



The Hate Of Pie Chart



https://medium.com/@clmentviguier/the-hate-of-pie-charts-harms-good-data-visualization-cc7cfed243b6

Avoid Rainbow Map

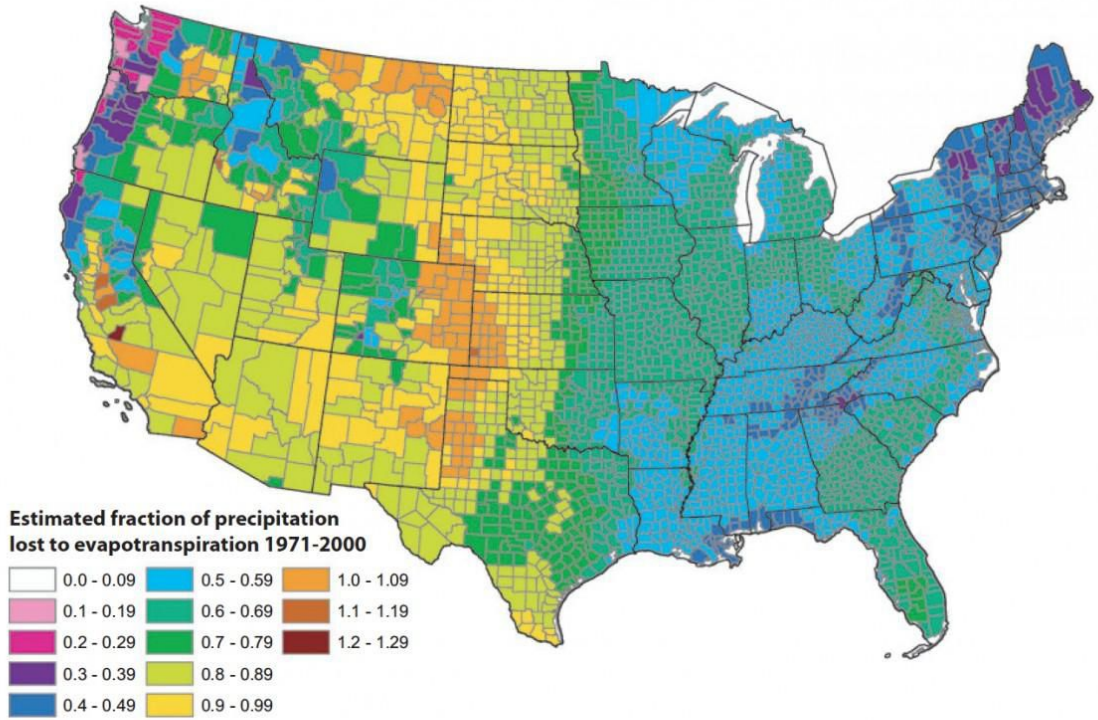


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

https://www.climate-lab-book.ac.uk/2016/why-rainbow-colour-scales-can-be-misleading/

Think About The Color Blind



Protanope Deuteranope Tritanope

Red / green deficiencies

Blue / Yellow deficiency

TRY!!! http://colorbrewer2.org/

https://cdn.mprog.nl/dataviz/slides/07-Color.pdf

Based on slide from Stone

LATCH

Information can only be organized in five possible ways (LATCH):

- Location
- Alphabet
- Time
- Category
- Hierarchy (or magnitude)

Location

- Geography or proximity, particularly when combined with time, is one of the most useful ways of organizing information.
- The location can be concrete, such as a position on a map, a part inside a system diagram, or a step in a flow chart, or more abstract directions of space, such as a grouping of images together to convey their relationship with the others (for example, French wine, Argentine beef, and so on).

Location



America Asia Europe

Alphabet

This is one of the less used in visual communication, but it provides a way to organize large amounts of data.

Although A fits nicely into its LATCH acronym, this dimension represents any scheme in which information is organized by an arbitrary symbol, such as a numerical index or table of contents (for example, phone book, list of websites, and so on).

Alphabetically



Afghan Boxer Collie Dachshund Greyhound Husky Poodle

Time

- Organizing things temporarily allows the suggestion of causality. This approach is most valuable when combined with other schemes such as location.
- Time is often represented by graphical schedules and calendars (for example, meeting of two hours, stays of two days, and so on).

Time



1930's 1950's 1970's

Category

- When you classify an item as part of a group, this can help you to identify patterns in your data. You can consider as examples of groups, distinctions such as gender, color, or other item attributes.
- Once the items are categorized, they can be counted and apply statistical techniques to identify those patterns (for example, format, topics, and so on).

Category



light color mixed color dark color

Hierarchy (or magnitude)

- The way items are related according to one or more factors will help you to find the internal structure and relationships between the data elements.
- For example, a hierarchy type would involve a simple item's attribute sorting, such as importance order, number, or its size.
- More complex hierarchies can be organizational charts, family trees, and other types of relational data (for example, price, size, and so on).

Hierarchy



small medium big

Following design trends

Keep in mind that colors communicate and provoke feelings

- Blue: The meaning of the color blue is mainly associated with the sensation of peace, as well as white, but in a more subtle way. It reminds you of cleanliness, water, serenity, and productivity. In dark tones it conveys safety, confidence, success, and power. It is used a lot in technology companies.
- Purple: The meaning of the color purple refers to wisdom, fantasy, mystery, and spirituality. Purple is a color that calms and imparts well-being, so beauty products, body care ads, and alternative treatments are frequent applications of this shade.
- Red: The symbolism of the color red represents love, emotion, excitement, romance, and pleasure. It also expresses urgency, and so is commonly used in the advertising of stores in clearance.
- Combined with yellow, it is widely used in food segments, as in restaurants and fast food chains, as it stimulates the appetite.
- Green: The color green is directly associated with health, nature, hope, and life. It reminds us of balance, freshness, harmony, and health. Its use in science, medicine, ecology, tourism, and organic food companies is quite common.

Keep in mind that colors communicate and provoke feelings

- Pink: A light pink color expresses innocence, while in darker tones it inspires desire, affection, and femininity. It is commonly used in products aimed at the female audience.
- Brown: The symbolism of the color brown conveys a sense of tradition, conservatism, reliability, solidity, and security, and it is widely used by furniture and interior decoration companies, and architecture.
- Yellow: The color yellow encourages creation and communication, and awakens joy, animation, and entertainment. For attention, it is used in traffic signs and also in store windows.
- Orange: The color orange represents an energetic, stimulating, young shade. It encourages expansion, creativity, enthusiasm, and optimism. It promotes change and dynamism. It is used in food, sports, and leisure segments.
- Black: The color black, according to the context, can represent sadness and mourning, but if it is used well it conveys nobility, tradition, curiosity, superiority, power, and professionalism. Engineering companies, law firms, cosmetics, and luxury products often use this color.
- White: The color white in combination with other colors is harmonious, expressing peace, faith, light, and purity. It is commonly used by medicine and dentistry companies.
- Gray: Gray is a classic, neutral color that conveys elegance and respect. It is used by technology and automotive

Color ads have a 62% higher visibility rate compared to black and white ads

- Red: Attracts impulsive shoppers, much used in fast foods and settlements Yellow: Aims to capture the attention of consumers in store windows, attracting impulsive consumers
- Blue: Widely used by corporations and large departments, can be seen in banks, seeking to bring a sense of security and confidence
- Green: Can be found in clinics and hospitals, also associated with money and wealth
- Orange: Aims to attract attention, also attracts impulsive buyers Purple: Associated with aesthetic clinics, beauty products, and fine environments, it attracts sophisticated consumers

Data Visualization

https://news.cgtn.com/event/2019/whorunschina/index.html

https://www.visualcinnamon.com/portfolio/why-do-cats-and-dogs/

A Best Practice

https://www.gapminder.org/

Q&A